Darshan University
School of Diploma Engineering

Mid Semester Examination, November- 2021
Sem.-I Examination (ALL BRANCH)

Enrollment. No.: _____

Subject: Mathematics-I (21HS02103) **Date: 22/11/2021**

Time : 11:00 am To 12:15 pm **Total Marks: 30**

Instructions: 1. Attempt all question
2. Assume suitable data if necessary
3. Only simple calculator allow.

Q – 1 **(a)** **1)** Simplify : $216x^3 - 343$ **2)** Simplify : $(2x - 3y - 4z)^2$ **02**

3) Simplify : $4x^2 - 3x - 10$ **4)** Simplify : $\frac{3^5 \times 10^5 \times 25}{5^7 \times 6^5}$

(b) **1)** Find the sum of the first 16th term of A.P.: 5 ,10, 15, … … .. **04**

2) If $\sin\theta = \frac{3}{5}$ then find all other trigonometry functions.

Q – 2 **(a)** **1)** $\log_3 162 - \log_3 2 = -----$ **2)** $\log_{\sqrt{3}}\left(\frac{1}{81}\right) = -------.$ **02**

3) $16^{-2\log_4 3} = ----$ **4)** $\log_{2\sqrt{2}}(256) = -----.$

(b) **1)** Simplify : $\frac{1}{\log_{p^2} pqr} + \frac{1}{\log_{r^2} pqr} + \frac{1}{\log_{q^2} pqr}$ **04**

2) Simplify : $\frac{1}{2}\log 25 + \frac{2}{3}\log 27 - \frac{1}{3}\log 64$

Q – 3 **(a)** **1)** Solve : $\log\left(x + \sqrt{x^2 - 1}\right) + \log\left(x - \sqrt{x^2 - 1}\right)$ **02**

2) Simplify : $\log_y \sqrt[3]{x} \times \log_z y^4 \times \log_x \sqrt[4]{z}$

(b) **1)** Solve : $\frac{\log x}{\log 8} = \frac{\log 256}{\log 64}$ **04**

2) If $\log\left(\frac{x-y}{2}\right) = 0.5\,(\log x + \log y)$ then P.T. $xy^{-1} + yx^{-1} = 6$.

Q – 4 **(a)** **1)** Evaluate : $\begin{vmatrix} 4 & 2 & 4 \\ -3 & -1 & -3 \\ 5 & 3 & 5 \end{vmatrix}$ **2)** If $\begin{vmatrix} 3 & -1 & 2 \\ 3 & x+1 & 2 \\ 1 & -5 & 4 \end{vmatrix} = 0$ then find x. **03**

3) If $A = \begin{bmatrix} -8 & -9 \\ -3 & -1 \end{bmatrix}$ then find $2A - 3I$.

(b) If $B = \begin{bmatrix} 4 & -6 & 8 \\ 8 & -4 & 6 \\ -6 & 8 & 4 \end{bmatrix}$, $C = \begin{bmatrix} -5 & 6 & -8 \\ -8 & -5 & 6 \\ 6 & -8 & 5 \end{bmatrix}$ & $A = \begin{bmatrix} -1 & -3 & -4 \\ -4 & -1 & 3 \\ 3 & 4 & -1 \end{bmatrix}$ **02**

Then find $2A - 3C + 4B$.

Q – 5 **(a)** **1)** If $A = \begin{bmatrix} -3 & -4 \\ -8 & -9 \\ -5 & -7 \end{bmatrix}$ then $A^T = ------$. **01**

2) _____ matrix is a 3 x 3 identity matrix.

(b) **1)** Evaluate : $\begin{vmatrix} e^{3x} & -e^{x} \\ -1 & e^{-2x} \end{vmatrix}$ **2)** If $\begin{vmatrix} 2b+3 & 5-3b \\ -1 & 4 \end{vmatrix} = -3$ then find b. **02**

(c) **1)** $A = \begin{vmatrix} 1 & -2 & 3 \\ -4 & 5 & -6 \\ 7 & -8 & 9 \end{vmatrix}$ Find all minors and cofactors given determinant. **04**

2) If $B = \begin{bmatrix} 2 & 3 & 4 \\ -8 & 9 & -1 \\ 1 & -3 & 4 \end{bmatrix}$ and $A = \begin{bmatrix} 4 & 6 & -8 \\ 2 & 3 & 4 \\ -5 & -8 & 7 \end{bmatrix}$ and given

Condition is $2(A - B) + 3C = 0$ then find matrix C.

ગુજરાતી

Q – 1 **(a)** **1)** સાદુંરૂપ આપો : $216x^3 - 343$ **2)** સાદુંરૂપ આપો : $(2x - 3y - 4z)^2$ **02**

3) સાદુંરૂપ આપો : $4x^2 - 3x - 10$ **4)** સાદુંરૂપ આપો : $\frac{3^5 \times 10^5 \times 25}{5^7 \times 6^5}$

(b) **1)** શ્રેણી $5, 10, 15, \ldots\ldots$ ના પ્રથમ 16 પદનો સરવાળો મેળવો. **04**

2) જો $\sin\theta = \frac{3}{5}$ હોય તો બાકીના ત્રિકોણમિતિય વિધેયો મેળવો.

Q – 2 **(a)** **1)** $\log_3 162 - \log_3 2 = ------$ **2)** $\log_{\sqrt{3}}\left(\frac{1}{81}\right) = -------$. **02**

3) $16^{-2\log_4 3} = ----$ **4)** $\log_{2\sqrt{2}}(256) = -----$.

(b) **1)** સાદુંરૂપ આપો : $\frac{1}{\log_{p^2} pqr} + \frac{1}{\log_{r^2} pqr} + \frac{1}{\log_{q^2} pqr}$ **04**

2) સાદુંરૂપ આપો : $\frac{1}{2}\log 25 + \frac{2}{3}\log 27 - \frac{1}{3}\log 64$

Q – 3 **(a)** **1)** ઉકેલો : $\log\left(x + \sqrt{x^2 - 1}\right) + \log\left(x - \sqrt{x^2 - 1}\right)$ **02**

2) સાદુંરૂપ આપો : $\log_y \sqrt[3]{x} \times \log_z y^4 \times \log_x \sqrt[4]{z}$

(b) **1)** ઉકેલો : $\frac{\log x}{\log 8} = \frac{\log 256}{\log 64}$ **04**

2) જો $\log\left(\frac{x-y}{2}\right) = 0.5\,(\log x + \log y)$ હોયતો સાબિતકરો $xy^{-1} + yx^{-1} = 6$.

Q – 4 **(a)** **1)** કિંમત મેળવો: $\begin{vmatrix} 4 & 2 & 4 \\ -3 & -1 & -3 \\ 5 & 3 & 5 \end{vmatrix}$ **2)** જો $\begin{vmatrix} 3 & -1 & 2 \\ 3 & x+1 & 2 \\ 1 & -5 & 4 \end{vmatrix} = 0$ હોય તો x મેળવો. **03**

3) જો $A = \begin{bmatrix} -8 & -9 \\ -3 & -1 \end{bmatrix}$ હોય તો $2A - 3I$ મેળવો.

(b) જો $B = \begin{bmatrix} 4 & -6 & 8 \\ 8 & -4 & 6 \\ -6 & 8 & 4 \end{bmatrix}$, $C = \begin{bmatrix} -5 & 6 & -8 \\ -8 & -5 & 6 \\ 6 & -8 & 5 \end{bmatrix}$ & $A = \begin{bmatrix} -1 & -3 & -4 \\ -4 & -1 & 3 \\ 3 & 4 & -1 \end{bmatrix}$ **02**

હોય તો $2A - 3C + 4B$ મેળવો.

Q – 5 **(a)** **1)** જો $A = \begin{bmatrix} -3 & -4 \\ -8 & -9 \\ -5 & -7 \end{bmatrix}$ હોય તો $A^T = ------$. **01**

2) _____ શ્રેણિક એ 3 x 3 એકમ શ્રેણિક છે.

(b) **1)** કિંમત મેળવો : $\begin{vmatrix} e^{3x} & -e^{x} \\ -1 & e^{-2x} \end{vmatrix}$ **2)** જો $\begin{vmatrix} 2b+3 & 5-3b \\ -1 & 4 \end{vmatrix} = -3$ હોય તો b મેળવો. **02**

(c) **1)** $A = \begin{vmatrix} 1 & -2 & 3 \\ -4 & 5 & -6 \\ 7 & -8 & 9 \end{vmatrix}$ ના બધા જ ઉપનિશ્ચાયક અને સહઅવયવ મેળવો. **04**

2) જો $B = \begin{bmatrix} 2 & 3 & 4 \\ -8 & 9 & -1 \\ 1 & -3 & 4 \end{bmatrix}$, $A = \begin{bmatrix} 4 & 6 & -8 \\ 2 & 3 & 4 \\ -5 & -8 & 7 \end{bmatrix}$ હોય અને

શરત $2(A - B) + 3C = 0$ હોય તો શ્રેણિક C મેળવો.

******BEST OF LUCK*******